The Augmented Synthetic Historical Control Method

1 Introduction

Applied causal inference in economic and policy research often compares a treated set of units to a control or donor pool. The synthetic control method (SCM) of Abadie et al. [2010] is one of the most widely used approaches in panel data econometrics, second only to difference-in-differences. The idea of SC is straightforward: the untreated potential outcomes of a treated unit are approximated by a weighted average of donor units. Difference-in-differences relies on the assumption of parallel pre-intervention trends [Baker et al., 2025].

In many settings, however, no suitable donor units are available—for example, during global events such as pandemics or financial crises, or when no valid donor units exist. When no valid untreated donor units are available, researchers often turn to the interrupted time series (ITS) design. In its canonical form, ITS models the pre-treatment outcome path as a parametric linear trend and tests for changes in level or slope following an intervention. In fact, even comparative interrupted time series designs have been used by policy analysts [Coopersmith et al., 2022]. Even so, ITS can be fragile in many realistic applications. Its validity depends critically on the assumption that an OLS line fitted to the pre-treatment period provides a reasonable counterfactual projection. In practice, many outcome series feature nonlinear dynamics, seasonality, or structural breaks that violate this assumption. Moreover, ITS typically imposes a homogeneous effect structure, assuming a single jump in level and/or slope that applies equally across all post-treatment periods. These restrictions can lead to biased inference when effects are time-varying or when pre-treatment is poor.

To address this shortcoming, Chen et al. [2024] proposed the synthetic historical control method (SHC). Rather than relying on external donors, SHC constructs a donor pool from the treated unit's own pre-treatment history, splitting the pre-treatment period into overlapping segments. The SHC also, unlike ITS, uses kernel smoothing to account for non-linearities in the pre-treatment time series data which ITS does not naturally account for. A quadratic program then estimates weights over these segments to predict counterfactual outcomes, allowing for effect size heterogeneity found in SCM and increasingly DID but not ITS. However, like Abadie et al. [2010, 2015], SHC restricts weights to the simplex. This guarantees interpretability but can be too restrictive when noise or complex dynamics make exact pre-treatment fit unattainable.

This paper introduces the Augmented Synthetic Historical Control method (ASHC), which extends SHC in two key ways. First, ASHC relaxes the simplex constraint to an affine constraint, requiring only that weights sum to one. This permits negative weights, allowing controlled extrapolation beyond the convex hull of observed segments in exchange for better fit. Second, ASHC adds a quadratic proximity penalty that discourages large deviations from the SHC solution, in the spirit of the augmented SCM [Ben-Michael et al., 2021]. This encourages the estimator to remain close to SHC unless extrapolation is needed. ASHC retains the stabilizing Gram penalty from Chen et al. [2024], which regularizes low-variance directions in the donor matrix. Together, these three components yield a new estimator that generalizes SHC to settings where exact pre-treatment fit is otherwise impossible.

The paper makes three contributions. Methodologically, it introduces the ASHC estimator, which generalizes SHC by permitting limited, penalized extrapolation while maintaining the stabilizing Gram penalty. Theoretically, it derives finite-sample error bounds that decompose ASHC prediction error into four interpretable parts: the SHC residual error, a regularization-controlled deviation term, smoothness-induced bias, and noise. It further establishes asymptotic consistency and a convergence rate of $O_p(m^{-H/(2H+3)})$, which approaches the near-parametric rate $O_p(m^{-1/2})$ as the smoothing order H increases. Practically, this means that when the pre-treatment length m is large (e.g. $m = 120$) and H grows, the convergence rate becomes practically close to the parametric $O_p(m^{-1/2})$, so ASHC achieves efficient performance in realistic sample sizes.

While the finite-sample analysis provides guarantees on the accuracy of ASHC for realistic data sizes, asymptotic results are equally important. The finite-sample bounds show how ASHC predictions may differ from SHC depending on regularization strength, operator norm, smoothness bias, and noise. The asymptotic results complement this by demonstrating that, as the amount of pre-treatment data grows, ASHC's prediction error vanishes at a near-parametric rate under mild conditions. Together, these perspectives ensure that ASHC is both practically reliable in finite samples and theoretically sound in large-sample limits.

The discussion begins with a careful review of SHC, its assumptions, and its implementation, before introducing the ASHC framework. We then turn to the theoretical analysis, establishing finite-sample error bounds and asymptotic properties. The empirical performance of ASHC is explored through calibrated simulations, followed by an application that illustrates its usefulness in practice. Proofs are collected in the appendix, and all computations are carried out using Python's mlsynth library (https://mlsynth.readthedocs.io/en/latest/).

For policy analysts, the message is straightforward: ASHC provides a principled way to construct reliable counterfactuals even in environments where no external donor units are available. Furthermore, it allows policy analysts to do this even when methods such as interrupted time series method or the Chen et al. [2024] would not do well.

The remainder of the paper proceeds as follows. Section 1.1 situates our contribution within the existing literature on causal inference in policy evaluation. We then review the SHC method in Section 2.1 before introducing the augmented SHC (ASHC) in Section 2.2. Section 3 presents our theoretical results, including finite-sample error bounds and asymptotic properties. We then turn to an empirical application in Section 4, replicating the analysis of Barcelona's 2015 hotel moratorium from Bel et al. [2021], a setting that allows us to compare ITS, SCM, SHC, and ASHC on equal footing. Finally, Section 5 concludes with implications for both methodological research and policy analysis.

1.1 Literature Review

This paper is situated within the literature on causal time series methods. The most popular method is likely the Bayesian Structural Time Series method [Brodersen et al., 2015, Dorsett, 2020, Tierney et al., 2024] due to its ease of use in Python and R as well as inferential methodology. Another similar estimator in the same vein is the causal ARIMA method [Menchetti et al., 2022].

The most natural setting this literature relates to is the artificial control setting with imperfect pre-treatment fit. Abadie et al. [2010, 2015] advise against using the SC method when pre-treatment fit is poor. This same kind of caution is implicit in the theoretical findings of Chen et al. [2024], who require a matching condition to hold for identification.

Econometricians are quite interested in how we may address this problem in realistic settings. Some researchers have argued for demeaning approaches [Ferman and Pinto, 2021] to remove fixed effects. Others advocate model averaging approaches [Kellogg et al., 2021], whereas still others study penalized approaches to use SCM when the normal method breaks down [Abadie and L'Hour, 2021]. This paper is of course most related to Ben-Michael et al. [2021] who argue for the affine hull and proximity penalty bias correction I use here for the SHC method. However of course, all the SCM methods require other units who are not treated, whereas SHC does not.

This paper is also within the fascinating recent work on doubly-robust models in panel data econometrics. Such designs have been proposed with SCM and DID style estimators, with validity under parallel trends or SCM's convex combination assumption [Sun et al., 2025, Petrillo et al., 2024, Arkhangelsky et al., 2024, Qiu et al., 2024]. Ben-Michael et al. [2021] makes the point that the ASCM (as well as ASHC) is not doubly robust in the formal sense, however, ASCM can be valid under certain conditions where SC attains imperfect pre-treatment fit.

2 Methodology

2.1 The Synthetic Historical Control Method

Let $\mathcal{T} = \{1, \ldots, T\}$ index time, with pre-treatment period $\mathcal{T}_1 = \{t \in \mathcal{T} : t \leq T_0\}$ and post-treatment period $\mathcal{T}_2 = \{t \in \mathcal{T} : t > T_0\}$. The number of post-treatment periods is $n = T - T_0$, and an evaluation window of length m is defined as the final m pre-treatment periods $\mathcal{T}_{\text{eval}} = \{T_0 - m + 1, \ldots, T_0\}$. Let $\mathbf{y} = (y_1, \ldots, y_T)^\top$ denote the observed outcome series. The evaluation subvector is $\mathbf{y}_{\text{eval}} = \mathbf{y}_{\mathcal{T}_{\text{eval}}} \in \mathbb{R}^m$, and the post-treatment outcome vector is $\mathbf{y}_{\text{post}} = \mathbf{y}_{\mathcal{T}_2} \in \mathbb{R}^n$. For each $t \in \mathcal{T}$, let y_t^1 and y_t^0 denote the potential outcomes under treatment and no treatment, respectively. The observed outcome is

$$y_t = d_t y_t^1 + (1 - d_t) y_t^0, \qquad d_t = \mathbf{1}\{t > T_0\}. \tag{2.1}$$

For $t > T_0$, only y_t^1 is observed. SHC provides an estimate $\widehat{y}_t^0$ for the missing counterfactual, and the treatment effect is estimated as

$$\widehat{\alpha}_t = y_t - \widehat{y}_t^0, \qquad t \in \mathcal{T}_2, \tag{2.2}$$

with the average treatment effect on the treated (ATT) defined as

$$\widehat{\text{ATT}} = \frac{1}{n} \sum_{t \in \mathcal{T}_2} \widehat{\alpha}_t. \tag{2.3}$$

The implementation of SHC begins by smoothing the pre-treatment outcome series $\mathbf{y}_{\text{pre}}$ using local linear regression with a Gaussian kernel, yielding a smooth trajectory $\widetilde{\mathbf{y}}$. From this smoothed series, we construct the donor matrix by extracting N overlapping segments, each of length m.

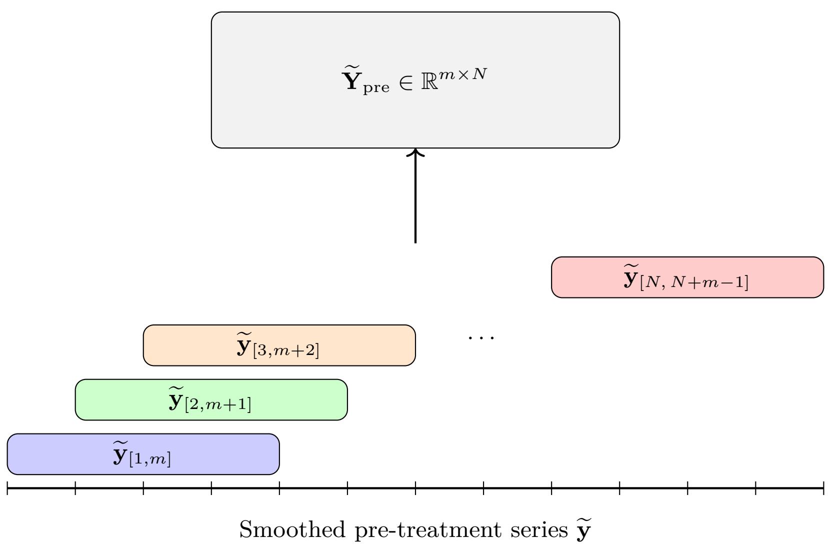

Figure 1: Constructing the donor matrix $\widetilde{\mathbf{Y}}_{\text{pre}}$ by sliding overlapping windows of length m across the smoothed pre-treatment series $\widetilde{\mathbf{y}}$. Each colored segment corresponds to a column in the donor matrix.

The evaluation vector $\mathbf{y}_{\text{eval}}$ is reconstructed as a convex combination of the donor segments by solving the quadratic program

$$\mathbf{w}^* = \arg\min_{\mathbf{w} \in \mathbb{R}^N} \left\| \mathbf{y}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{pre}} \mathbf{w} \right\|_2^2 + \varsigma \left\| \mathbf{C}_0^\top \mathbf{w} \right\|_2^2, \qquad \text{subject to } \mathbf{w} \geq \mathbf{0},\ \mathbf{1}^\top \mathbf{w} = 1, \tag{2.4}$$

where $\mathbf{C}_0$ contains the eigenvectors of the donor Gram matrix associated with eigenvalues below a threshold τ. The additional penalty shrinks weights along low-variance directions, mitigating instability from collinearity. Applying $\mathbf{w}^*$ to the aligned post-treatment donor matrix yields $\widehat{\mathbf{y}}_{\text{post}}^0$, which serves as the SHC estimate of the untreated counterfactual trajectory.

2.2 The Augmented Synthetic Historical Control Method

The augmented synthetic historical control (ASHC) method extends SHC by permitting limited extrapolation beyond the convex hull of historical segments, while stabilizing the solution through a proximity penalty. This modification is motivated by cases where the simplex constraint of SHC is too restrictive to achieve adequate pre-treatment fit. By allowing negative weights under an affine constraint and penalizing deviations from the SHC solution, ASHC balances improved fit with regularization. Let $\mathbf{w}^{\mathrm{SHC}}$ denote the SHC weight vector. The ASHC estimator solves

$$\mathbf{w}^{\mathrm{ASHC}} = \arg\min_{\mathbf{w}\in\mathbb{R}^N} \underbrace{\left\|\mathbf{y}_{\mathrm{eval}} - \widetilde{\mathbf{Y}}_{\mathrm{pre}}\mathbf{w}\right\|_2^2}_{\text{Fit Term}} + \underbrace{\varsigma\left\|\mathbf{C}_0^\top\mathbf{w}\right\|_2^2}_{\text{Low-variance term}} + \underbrace{\lambda\left\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\right\|_2^2}_{\text{Proximity term}}, \qquad \text{subject to } \mathbf{1}^\top\mathbf{w} = 1. \quad (2.5)$$

As we can see, the first two terms are essentially unchanged from the original SHC method. The third term in the ASHC objective,

$$\lambda\left\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\right\|_2^2,$$

is a proximity penalty. This term discourages the ASHC solution from deviating too far from the original SHC weights. The penalty plays a dual role. Conceptually, it regularizes the affine extension of SHC by shrinking the solution back toward the purely convex combination used in SHC. However, the additional flexibility provided by negative weights is only exercised when it leads to a substantial reduction in pre-treatment error. The tuning parameter $\lambda \geq 0$ controls the degree of shrinkage: when λ is very large, the ASHC solution converges to $\mathbf{w}^{\mathrm{SHC}}$, while when $\lambda = 0$ the penalty vanishes and the estimator reduces to an unpenalized affine extension of SHC. Thus, the proximity penalty creates a continuum between SHC and its affine relaxation, allowing researchers to balance interpretability and stability against the potential gains in fit from extrapolation. The ASHC counterfactual is obtained as $\widehat{\mathbf{y}}_{\mathrm{post}}^{0,\mathrm{ASHC}} = \widetilde{\mathbf{Y}}_{\mathrm{post}}\,\mathbf{w}^{\mathrm{ASHC}}$, with treatment effects estimated analogously to SHC.

2.3 Bandwidth and Regularization Parameter Selection

The performance of ASHC depends critically on two hyperparameters: the smoothing bandwidth h and the proximity penalty parameter λ. Both are chosen via cross-validation. First I discuss the bandwidth for the kernel smoothing. Let $\mathbf{y}_{\mathrm{pre}} = (y_1, \ldots, y_{T_0})^\top$ denote the pre-treatment series. For a candidate bandwidth $h > 0$, define the local linear smoother estimate $\widehat{y}_i^{(-i)}(h)$ as the fitted intercept obtained by regressing $\{y_j : j \neq i\}$ on $\{j - i : j \neq i\}$ with kernel weights

$$K_h(j-i) \;=\; \exp\Big(-\tfrac{1}{2}\left(\tfrac{j-i}{h}\right)^2\Big),$$

normalized so that $\sum_{j\neq i} K_h(j-i) = 1$. Formally, for each $i \in \{1, \ldots, T_0\}$,

$$\widehat{\beta}(h)^{(-i)} = \arg\min_{\beta_0,\beta_1} \sum_{j\neq i} K_h(j-i)\left(y_j - \beta_0 - \beta_1(j-i)\right)^2,$$

and the leave-one-out prediction is $\widehat{y}_i^{(-i)}(h) = \widehat{\beta}_0(h)^{(-i)}$. The LOOCV error is $\mathrm{CV}(h) = \frac{1}{T_0}\sum_{i=1}^{T_0}\Big(y_i - \widehat{y}_i^{(-i)}(h)\Big)^2$. The optimal bandwidth is then chosen as $h^* = \arg\min_{h\in\mathcal{H}} \mathrm{CV}(h)$, where $\mathcal{H}$ is a prespecified candidate grid. This is part of the normal SHC method.

Next we discuss λ, exclusive to the ASHC. Recall the ASHC objective

$$Q(\mathbf{w};\lambda) = \left\|\mathbf{y}_{\mathrm{train}} - \widetilde{\mathbf{Y}}_{\mathrm{train}}\mathbf{w}\right\|_2^2 + \varsigma\left\|\mathbf{C}_0^\top\mathbf{w}\right\|_2^2 + \lambda\left\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\right\|_2^2, \qquad \text{subject to } \mathbf{1}^\top\mathbf{w} = 1. \qquad (2.6)$$

Let the evaluation window $\mathcal{T}_{\mathrm{eval}}$ be split into a training set $\mathcal{T}_{\mathrm{train}}$ and a validation set $\mathcal{T}_{\mathrm{val}}$, with corresponding subvectors $\mathbf{y}_{\mathrm{train}}, \mathbf{y}_{\mathrm{val}}$ and donor matrices $\widetilde{\mathbf{Y}}_{\mathrm{train}}, \widetilde{\mathbf{Y}}_{\mathrm{val}}$.

We construct a logarithmically spaced grid of candidate values

$$\Lambda = \left\{\lambda_j \ : \ \lambda_j \ = \ \lambda_{\max} \, \rho^{j}, \ \ j = 0, 1, \ldots, n_\lambda - 1\right\}, \tag{2.7}$$

where

$$\lambda_{\max} \ = \ \sigma^2_{\max}, \qquad \lambda_{\min} \ = \ \lambda_{\max} \cdot \kappa, \qquad \rho \ = \ \left(\frac{\lambda_{\min}}{\lambda_{\max}}\right)^{\frac{1}{n_\lambda - 1}}. \tag{2.8}$$

Here $\sigma_{\max}$ denotes the largest singular value of $\widetilde{\mathbf{Y}}_{\text{train}}$, and $\kappa > 0$ is a user-chosen constant controlling the lower bound of the grid.[1] Thus Λ spans values from $\lambda_{\max}$ (very strong shrinkage toward $\mathbf{w}^{\text{SHC}}$) down to $\lambda_{\min}$ (minimal shrinkage).

For each $\lambda \in \Lambda$, the estimator is

$$\widehat{\mathbf{w}}(\lambda) = \arg\min_{\mathbf{w} \in \mathbb{R}^N} Q(\mathbf{w}; \lambda), \tag{2.9}$$

and the validation error is

$$\text{Val}(\lambda) = \frac{1}{|\mathcal{T}_{\text{val}}|} \left\|\mathbf{y}_{\text{val}} - \widetilde{\mathbf{Y}}_{\text{val}} \, \widehat{\mathbf{w}}(\lambda)\right\|_2^2. \tag{2.10}$$

The optimal penalty parameter is selected as

$$\lambda^* = \arg\min_{\lambda \in \Lambda} \text{Val}(\lambda). \tag{2.11}$$

Together, h^* and λ^* balance the dual bias–variance tradeoffs of ASHC: smoothing bias versus noise, and extrapolation bias versus instability.

3 Theoretical Results for Finite and Asymptotic Settings

I establish the finite-sample guarantees and asymptotic properties of ASHC. We work within the fixed N, large T asymptotic framework. My analysis proceeds in three steps: First, I bound how far ASHC weights can deviate from the SHC weights. Then, I derive finite-sample error bounds, first for the pre-treatment period and then for the post-treatment period. Finally, I establish asymptotic consistency and the convergence rate as the evaluation window grows. Complete proofs are deferred to the Appendix.

Chen et al. [2024] make the following assumptions for SHC's validity.

Assumption 3.1 (Error Properties)**.**

$$\mathbb{E}[\varepsilon_t] = 0 \quad \textit{and} \quad \mathbb{E}[\varepsilon_t^2] < \infty \qquad \forall\, t \in \mathcal{T}. \tag{3.12}$$

This assumption ensures that noise is well-behaved. The mean-zero condition rules out systematic bias from the error term, while the finite variance condition prevents extreme fluctuations. Together, these conditions guarantee that the noise does not overwhelm the signal we seek to estimate.

Assumption 3.2 (Latent Smoothness and Matching)**.**

$$\widetilde{y}(t) \in C^{H+1}, \quad H \geq 3, \qquad \sup_t \left|\frac{d^k}{dt^k}\widetilde{y}(t)\right| < \infty \ \ \forall\, k \leq H+1, \tag{3.13}$$

and there exists

$$\mathbf{w}^* \in \Delta^{N-1} \qquad \textit{such that} \qquad \mathbf{y}_{eval} = \widetilde{\mathbf{Y}}_{pre}\mathbf{w}^*. \tag{3.14}$$

This assumption serves two purposes. First, differentiability ensures that the latent outcome process is smooth, so it can be approximated by a finite harmonic expansion. Second, the matching condition guarantees that there exists some convex combination of donor segments that can reproduce the evaluation period exactly, at least in the absence of noise.

[1] In practice, we set $\kappa = 10^{-8}$, ensuring that the grid spans from strong shrinkage ($\lambda_{\max}$) to nearly unpenalized fits ($\lambda_{\min}$).

Assumption 3.3 (Feasibility and Stability)**.**

$$\exists\, \mathbf{w} \in \Delta^{N-1} \;\; \textit{sparse, such that} \;\; \mathbf{y}_{eval} \approx \widetilde{\mathbf{Y}}_{pre}\mathbf{w}, \qquad \widetilde{\mathbf{Y}}_{pre}^{\top}\widetilde{\mathbf{Y}}_{pre} \succ 0. \tag{3.15}$$

This assumption ensures that a stable solution exists. Sparsity reflects the idea that only a limited number of donor segments are truly relevant. The positive definiteness of the Gram matrix prevents collinearity among donors, which could otherwise lead to unstable weight estimates. I make two additional assumptions for ASHC.

Assumption 3.4 (Operator Norm Bound)**.**

$$\|\widetilde{\mathbf{Y}}\| \;\leq\; \gamma, \qquad \textit{for some constant } \gamma > 0. \tag{3.16}$$

This assumption restricts the size of the donor matrix in operator norm. Intuitively, it ensures that small deviations in weights cannot translate into disproportionately large changes in predicted outcomes. Without this bound, the affine extension in ASHC could magnify minor instabilities in the estimation.

Assumption 3.5 (Post-Treatment Operator Norm)**.**

$$\|\widetilde{\mathbf{Y}}_{post}\| \;\leq\; \gamma_{post}, \qquad \textit{for some constant } \gamma_{post} > 0. \tag{3.17}$$

This is the post-treatment analogue of Assumption~3.4. It ensures that when the ASHC weights are applied to post-treatment donor segments, the resulting counterfactual predictions remain stable and well-controlled. Together, these two operator norm bounds protect against instability both before and after treatment.

To study the finite sample properties of ASHC, we first must bound the change in the ASHC weights. We do begin using Lemma 3.1. Let $\Delta := \mathbf{w}^{\text{ASHC}} - \mathbf{w}^{\text{SHC}}$ denote the deviation of ASHC weights from the SHC solution (i.e., will not explode too much).

Lemma 3.1 (Deviation from SHC)**.** *Under Assumptions 3.1–3.4,*

$$\|\Delta\| \;\leq \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\, \epsilon_{SHC}, \tag{3.18}$$

where $\epsilon_{SHC} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ is the SHC in-sample fit error.

This Lemma means that even if ASHC's weights differ from SHC's, the resulting change in fitted donor outcomes is bounded by the donor matrix's operator norm. If the donor matrix were too "large," small weight deviations could produce implausibly large changes in predictions. Bounding the operator norm guarantees that such deviations remain controlled.

Theorem 3.1 (Finite-Sample Error Bound for ASHC)**.** *Under Assumptions 3.1–3.4, the ASHC estimator satisfies*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \leq \epsilon_{SHC}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_{\epsilon}(H,k) + \delta, \tag{3.19}$$

where $\epsilon_{SHC} = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{SHC}\|$ is the in-sample SHC fit error, $b_{\epsilon}(H,k)$ is the smoothness bias from the smoothing approximation, δ is the noise bias, and $\gamma, \lambda, \varsigma$ are constants defined in the model.

Remark 3.1. *This provides a pre-treatment error bound that decomposes into four components. The first is the baseline error, which is simply the in-sample fit error of the standard SHC model, ϵ_{SHC}. The second component is a deviation penalty, controlled by the regularization parameters λ and ς. This term quantifies the additional error introduced by allowing the ASHC weights to deviate from the SHC weights, such that this deviation doesn't lead to a significant loss of fit. The third component, $b_{\epsilon}(H,k)$, represents the smoothness bias that arises from the Taylor approximation used in smoothing step. This bias is a function of the number of harmonics, H, and the order of the approximation, k, and reflects the error introduced by approximating a smooth function (which may be less approximated with noisier outcome series). The final component, δ, is the bias from random noise, which accounts for the unpredictable, random fluctuations in the data that are not explained by the model.*

Similar bounds exist for the ATT estimate. This is shown in Prosperous Proposition 3.1, which places am upper bound in the bias of the ATT.

Proposition 3.1 (Out-of-sample Error Bound)**.** *Under Assumptions 3.1–3.5,*

$$\left\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^{ASHC}\right\| \leq b_{\epsilon,post}(H,k) + \delta_{post} + \gamma_{post}\left(\eta + \frac{\epsilon_{SHC}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right), \tag{3.20}$$

where η *bounds* $\|\mathbf{w}^* - \mathbf{w}^{SHC}\|$.

This clarifies the factors influencing the accuracy of ASHC's ATT estimates. The bound consists of four key components. The first is post-treatment smoothness bias, $b_{\epsilon,\text{post}}(H,k)$, which is the error from using a spherical harmonic approximation to model the post-treatment outcome trend. The second is post-treatment noise, δ_{post}, which captures random error in the outcome data. The third component, ϵ_{SHC}, represents the quality of the initial SHC fit and demonstrates that a poor pre-treatment fit can propagate into post-treatment error. This term is scaled by the operator norm, γ_{post}, and regularization parameters, λ and ς, highlighting how regularization helps to control the influence of the initial fit error. The final term, η, is a bound on the difference between the optimal weights and the SHC weights, which is also scaled by γ_{post}. This term represents the bias from regularization itself.

We now turn to large-sample guarantees. While finite-sample performance is most critical in practice, since most SCM/SHC based methods do not have very large sample sizes compared to DID, ASHC remains theoretically reliable in large-sample limits.

The asymptotic results show that as the number of pre-treatment periods grows, the ASHC estimator becomes consistent: its prediction error shrinks to the noise level, and under mild sub-Gaussian error conditions, converges in probability to zero. Theorem 3.2 provides a consistency result as the pre-treatment period grows without bound.

Theorem 3.2 (Asymptotic Consistency of ASHC)**.** *Under Assumptions 3.1–3.5, as* $m \to \infty$ *and* $H \to \infty$*, the ASHC estimator satisfies*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{ASHC}\| \xrightarrow{p} 0. \tag{3.21}$$

While $\boldsymbol{\eta}$ *is sub-Gaussian, then* $\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| \xrightarrow{p} 0$.

Remark 3.2. *With enough pre-treatment data and smooth latent trends, ASHC's prediction error shrinks to the noise level, and under mild probabilistic conditions converges in probability to zero. Note that bias will never vanish completely due to the smoothing terms. However, in practice, this means longer evaluation windows and pre-treatment periods yield increasingly unbiased out of sample predictions.*

With the asymptotic convergence established, we can derive the rate too.

Corollary 3.1 (Convergence Rate)**.** *If* $\boldsymbol{\eta}$ *is i.i.d. mean-zero sub-Gaussian with variance proxy* σ^2 *and bandwidth* $h_m \asymp m^{-1/(2H+3)}$*, then as* $m \to \infty$*,*

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{ASHC}\| = O_p\left(m^{-H/(2H+3)}\right). \tag{3.22}$$

For large H*, this approaches the parametric rate* $O_p(m^{-1/2})$.

The expression $O_p\big(m^{-H/(2H+3)}\big)$ describes the convergence rate of the ASHC estimator's prediction error. The term $O_p(\cdot)$ is the "Big O in probability" notation, which describes how quickly a sequence of random variables, in this case the prediction error, converges to zero as a parameter approaches infinity. The parameter m represents the number of pre-treatment data points (the final m periods before the treatment), while H is the number of spherical harmonics used to model the underlying latent trend, with a larger H corresponding to a smoother function. The ASHC estimator's prediction error decreases at a rate proportional to $m^{-H/(2H+3)}$. The negative exponent means that as the amount of evaluation data (m) increases, the prediction error decreases. A key insight is that the convergence rate depends on H; as H increases, the exponent $-H/(2H+3)$ becomes larger in magnitude, causing the error to decrease more rapidly. For very large H, this rate approaches the parametric rate of $O_p(m^{-1/2})$, which is the fastest possible convergence rate for an estimator.

The ASHC method offers a practical solution for policy analysis. It achieves better pre-treatment balance where the standard SHC method might fall short, without compromising the model's stability or interpretability. Also, it builds counterfactuals without the need for a donor set of units. This allows for credible counterfactual estimates in situations without donor units and standard ITS approaches fail. Moreover, the SHC methods are semiparametric, meaning they do not rely on the same more rigid assumptions on the data generating process oftentimes found in linear factor models or latent variable models.

4 Empirical Application

We replicate the setting of Bel et al. [2021], who study the effects of Barcelona's 2015 hotel moratorium using data on hotel prices from 2010 to 2017. This application provides a compelling case to benchmark ITS, SCM, SHC, and ASHC against one another.

5 Conclusion

Appendix

Mathematical Proofs

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A Useful Lemmas

Proof of Lemma 3.1. To analyze the ASHC optimization problem, recall the objective function:

$$J(\mathbf{w}) = \frac{1}{2\lambda}\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}\|^2 + \|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 + \varsigma\|\mathbf{C}_0^\top \mathbf{w}\|^2, \tag{1.23}$$

subject to the affine constraint $\mathbf{1}^\top \mathbf{w} = 1$.

To enforce this constraint, we form th Lagrangian:

$$\mathcal{L}(\mathbf{w}, \mu) = J(\mathbf{w}) + \mu(\mathbf{1}^\top \mathbf{w} - 1), \tag{1.24}$$

where μ is a Lagrange multiplier ensuring feasibility of $\mathbf{w}$. We compute the gradient of $\mathcal{L}$ with respect to $\mathbf{w}$ by differentiating each term separately.

For the first term, define the residual vector:

$$r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}. \tag{1.25}$$

The fit term is then:

$$\frac{1}{2\lambda}\|r(\mathbf{w})\|^2 = \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w}). \tag{1.26}$$

We can see that this is a standard quadratic form. We begin by applying the Chain Rule for vector calculus. The chain rule for a composite function $f(g(\mathbf{w}))$ is given by:

$$\nabla_{\mathbf{w}} f(g(\mathbf{w})) = \left(\nabla_{g(\mathbf{w})} f\right)^\top \nabla_{\mathbf{w}} g(\mathbf{w})$$

In our case, the inner function is the residual vector $g(\mathbf{w}) = r(\mathbf{w})$, and the outer function is the squared norm $f(r) = \frac{1}{2\lambda} r^\top r$. First, let's find the gradient of the outer function with respect to the residual vector r.

$$\nabla_r \left(\frac{1}{2\lambda} r^\top r\right) = \frac{1}{2\lambda} \nabla_r \left(\sum_i r_i^2\right) = \frac{1}{2\lambda} \begin{pmatrix} 2r_1 \\ 2r_2 \\ \vdots \end{pmatrix} = \frac{1}{\lambda} r$$

This is a well-known result: the gradient of $\|r\|^2$ is $2r$. The $\frac{1}{2\lambda}$ is a constant scaling factor.

Next, we need to find the gradient of the inner function, $r(\mathbf{w})$, with respect to the weight vector $\mathbf{w}$. This is the Jacobian matrix of the transformation.

$$\nabla_{\mathbf{w}} r(\mathbf{w}) = \nabla_{\mathbf{w}} (\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Since $\widetilde{\mathbf{y}}$ is a constant vector with respect to $\mathbf{w}$, its gradient is zero. We only need to compute the gradient of $-\widetilde{\mathbf{Y}}\mathbf{w}$.

$$\nabla_{\mathbf{w}} (-\widetilde{\mathbf{Y}}\mathbf{w}) = -\widetilde{\mathbf{Y}}$$

This result comes from the general rule that for a linear function $A\mathbf{x}$, the gradient with respect to $\mathbf{x}$ is just the matrix A. Here, our linear function is a negative, so the gradient is $-\widetilde{\mathbf{Y}}$.

Now we plug the results from the above back into the chain rule formula.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\nabla_{r(\mathbf{w})} \frac{1}{2\lambda} r(\mathbf{w})^\top r(\mathbf{w})\right)^\top \nabla_{\mathbf{w}} r(\mathbf{w})$$

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \left(\frac{1}{\lambda} r(\mathbf{w})\right)^\top (-\widetilde{\mathbf{Y}})$$

Using the property that $(A\mathbf{x})^\top = \mathbf{x}^\top A^\top$, we get:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda} r(\mathbf{w})^\top (-\widetilde{\mathbf{Y}}) = -\frac{1}{\lambda} r(\mathbf{w})^\top \widetilde{\mathbf{Y}}$$

This is equivalent to:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top r(\mathbf{w})$$

Finally, we substitute the definition of the residual vector, $r(\mathbf{w}) = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}$, back into the gradient expression.

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = -\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w})$$

Distributing the negative sign gives us the final form:

$$\nabla_{\mathbf{w}} J(\mathbf{w}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}})$$

Next, consider the deviation penalty term:

$$\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = (\mathbf{w} - \mathbf{w}^{\mathrm{SHC}})^\top(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.27}$$

Its gradient with respect to $\mathbf{w}$ is:

$$\nabla_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}). \tag{1.28}$$

Notice that the Hessian of this term is $2\mathbf{I}$, because differentiating again with respect to $\mathbf{w}$ gives:

$$\nabla^2_{\mathbf{w}}\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2 = 2\mathbf{I}. \tag{1.29}$$

Here $\mathbf{I}$ is the identity matrix, representing curvature equally in all coordinate directions. Thus the $2\mathbf{I}$ in M directly reflects the Hessian of this quadratic deviation penalty.

For the Gram penalty term, rewrite it as:

$$\varsigma\|\mathbf{C}_0^\top\mathbf{w}\|^2 = \varsigma\mathbf{w}^\top\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}. \tag{1.30}$$

Since $\mathbf{C}_0\mathbf{C}_0^\top$ is symmetric, its gradient is:

$$\nabla_{\mathbf{w}}\varsigma\mathbf{w}^\top\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} = 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}. \tag{1.31}$$

Finally, the affine constraint term differentiates as:

$$\nabla_{\mathbf{w}}\mu(\mathbf{1}^\top\mathbf{w} - 1) = \mu\mathbf{1}. \tag{1.32}$$

We now have the first-order condition from the gradient of the Lagrangian:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top(\widetilde{\mathbf{Y}}\mathbf{w} - \widetilde{\mathbf{y}}) + 2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.33}$$

Expanding the deviation penalty term gives

$$2(\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}) = 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}}. \tag{1.34}$$

Substituting back, the condition becomes

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w} - \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w} - 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w} + \mu\mathbf{1} = \mathbf{0}. \tag{1.35}$$

Now collect all the terms involving $\mathbf{w}$ on the left-hand side:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} \tag{1.36}$$

and move all the other terms to the right-hand side:

$$\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.37}$$

Thus we obtain the compact matrix form:

$$\left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\right)\mathbf{w} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.38}$$

Here the $2\mathbf{I}$ arises directly from the quadratic penalty $\|\mathbf{w} - \mathbf{w}^{\mathrm{SHC}}\|^2$, whose gradient contributes $2\mathbf{w}$, corresponding in matrix form to $2\mathbf{I}\cdot\mathbf{w}$.

Next, define the matrix

$$M = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top. \tag{1.39}$$

M corresponds to the Hessian of the objective function $J(\mathbf{w})$ plus the identity-weighted contribution of the deviation penalty.

The solution $\mathbf{w}^{\mathrm{ASHC}}$ satisfies the first-order condition:

$$M\mathbf{w}^{\mathrm{ASHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}. \tag{1.40}$$

Evaluating the same matrix M at $\mathbf{w}^{\mathrm{SHC}}$, which is not necessarily the ASHC solution, gives:

$$M\mathbf{w}^{\mathrm{SHC}} = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}. \tag{1.41}$$

Subtracting this equation from the previous one, we get

$$M\mathbf{w}^{\mathrm{ASHC}} - M\mathbf{w}^{\mathrm{SHC}} = \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{y}} + 2\mathbf{w}^{\mathrm{SHC}} - \mu\mathbf{1}\right) - \left(\frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + 2\mathbf{w}^{\mathrm{SHC}} + 2\varsigma\mathbf{C}_0\mathbf{C}_0^\top\mathbf{w}^{\mathrm{SHC}}\right). \tag{1.42}$$

Simplifying the right-hand side by canceling the $2\mathbf{w}^{\mathrm{SHC}}$ terms, we obtain

$$M(\mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}) = \frac{1}{\lambda}\widetilde{\mathbf{Y}}^\top\left(\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\right) - \mu\mathbf{1}. \tag{1.43}$$

Define the deviation

$$\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}, \tag{1.44}$$

and the residual vector

$$r = \widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}. \tag{1.45}$$

Since both $\mathbf{w}^{\mathrm{ASHC}}$ and $\mathbf{w}^{\mathrm{SHC}}$ satisfy the affine constraint $\mathbf{1}^\top\mathbf{w} = 1$, their difference Δ lies in the hyperplane orthogonal to $\mathbf{1}$:

$$\mathbf{1}^\top\Delta = 0. \tag{1.46}$$

Taking the inner product of both sides of the previous equation with Δ gives

$$\Delta^\top M\Delta = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r - \mu\underbrace{\Delta^\top\mathbf{1}}_{=0} = \frac{1}{\lambda}\Delta^\top\widetilde{\mathbf{Y}}^\top r. \tag{1.47}$$

Here, the right-hand side is an inner product between the deviation vector Δ and the vector $\widetilde{\mathbf{Y}}^\top r$. Because inner products can be difficult to control directly, we apply the Cauchy–Schwarz inequality to bound it:

$$|\Delta^\top\widetilde{\mathbf{Y}}^\top r| \le \|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|. \tag{1.48}$$

Thus, $\Delta^\top M\Delta \le \frac{1}{\lambda}\|\Delta\|\,\|\widetilde{\mathbf{Y}}^\top r\|$. By assumption, the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, and the residual norm is $\epsilon_{\mathrm{SHC}} = \|r\|$, so

$$\|\widetilde{\mathbf{Y}}^\top r\| \le \|\widetilde{\mathbf{Y}}\|\|r\| \le \gamma\epsilon_{\mathrm{SHC}}. \tag{1.49}$$

Thus, $\Delta^\top M\Delta \le \frac{\gamma}{\lambda}\|\Delta\|\epsilon_{\mathrm{SHC}}$. On the other hand, since M is positive definite, its smallest eigenvalue $\lambda_{\min}(M)$ guarantees

$$\Delta^\top M\Delta \;\ge\; \lambda_{\min}(M)\|\Delta\|^2. \tag{1.50}$$

To see this, recall the Rayleigh quotient characterization of eigenvalues:

$$\lambda_{\min}(M) \;=\; \min_{\mathbf{z}\neq 0}\frac{\mathbf{z}^\top M\mathbf{z}}{\|\mathbf{z}\|^2}. \tag{1.51}$$

By setting $\mathbf{z} = \Delta$, we obtain

$$\frac{\Delta^\top M\Delta}{\|\Delta\|^2} \;\ge\; \lambda_{\min}(M), \tag{1.52}$$

which rearranges to the inequality above.

Putting the two inequalities together, we sandwich the quadratic form:

$$\lambda_{\min}(M)\|\Delta\|^2 \;\le\; \Delta^\top M\Delta \;\le\; \frac{\gamma}{\lambda}\|\Delta\|\,\epsilon_{\mathrm{SHC}}. \tag{1.53}$$

Assuming $\|\Delta\| > 0$ (the trivial case where the weights coincide is clear), we divide through by $\|\Delta\|$ to obtain:

$$\lambda_{\min}(M)\,\|\Delta\| \;\le\; \frac{\gamma}{\lambda}\,\epsilon_{\mathrm{SHC}}. \tag{1.54}$$

Rearranging yields the intermediate bound:

$$\|\Delta\| \;\le\; \frac{\gamma}{\lambda\,\lambda_{\min}(M)}\,\epsilon_{\mathrm{SHC}}. \tag{1.55}$$

Next, we establish a lower bound for $\lambda_{\min}(M)$. Recall that

$$M \;=\; \frac{1}{\lambda}\,\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}} + 2\mathbf{I} + 2\varsigma\,\mathbf{C}_0\mathbf{C}_0^\top. \tag{1.56}$$

By the variational characterization of eigenvalues, the smallest eigenvalue of a sum of symmetric positive semidefinite matrices is at least the sum of their individual smallest eigenvalues:

$$\lambda_{\min}(M) \;\ge\; \lambda_{\min}\Big(\tfrac{1}{\lambda}\,\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}\Big) + \lambda_{\min}(2\mathbf{I}) + \lambda_{\min}(2\varsigma\,\mathbf{C}_0\mathbf{C}_0^\top). \tag{1.57}$$

Since $\widetilde{\mathbf{Y}}^\top\widetilde{\mathbf{Y}}/\lambda$ is positive semidefinite, its smallest eigenvalue is nonnegative, and we know $\lambda_{\min}(2\mathbf{I}) = 2$. Under Assumption 3.3, $\lambda_{\min}(\mathbf{C}_0\mathbf{C}_0^\top) \ge \|\mathbf{C}_0\|^2$, so

$$\lambda_{\min}(M) \;\ge\; 0 + 2 + 2\varsigma\,\|\mathbf{C}_0\|^2. \tag{1.58}$$

Substituting this bound into the inequality for $\|\Delta\|$ gives the final deviation bound:

$$\boxed{\|\Delta\| \;\le\; \frac{\gamma}{2\lambda + 2\varsigma\,\|\mathbf{C}_0\|^2}\,\epsilon_{\mathrm{SHC}}} \tag{1.59}$$

This result shows that the deviation between the ASHC and SHC weights is directly proportional to the SHC residual error, and shrinks as either λ (the proximity penalty) or ς (the Gram stability penalty) increases. ☐

Lemma 1.2. *Under Assumption (A4), the error from deviating off SHC weights satisfies*

$$\|\widetilde{\mathbf{Y}}\Delta\| \le \gamma \|\Delta\|,$$

where $\Delta = \mathbf{w}^{ASHC} - \mathbf{w}^{SHC}$.

Proof of Lemma 1.2. We begin by recalling the definition of the operator (spectral) norm of a matrix. For a real matrix $A \in \mathbb{R}^{m\times N}$,

$$\|A\| \ := \sup_{\mathbf{v}\in\mathbb{R}^N\setminus\{0\}} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This means $\|A\|$ measures the largest possible multiplicative factor by which A can increase the Euclidean norm of a vector.

Now let $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, where $\Delta = \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}} \in \mathbb{R}^N$ is the deviation of the ASHC weight vector from the SHC weight vector.

By the definition of the operator norm, we have

$$\frac{\|\widetilde{\mathbf{Y}}\Delta\|}{\|\Delta\|} \ \le \ \|\widetilde{\mathbf{Y}}\|, \quad \text{for all } \Delta \neq \mathbf{0}.$$

Multiplying through by $\|\Delta\|$ yields

$$\|\widetilde{\mathbf{Y}}\Delta\| \ \le \ \|\widetilde{\mathbf{Y}}\|\,\|\Delta\|.$$

Finally, Assumption (A4) gives us the explicit bound

$$\|\widetilde{\mathbf{Y}}\| \ \le \ \gamma,$$

so that

$$\|\widetilde{\mathbf{Y}}\Delta\| \ \le \ \gamma\|\Delta\|.$$

In words: although ASHC may choose weights that differ from SHC, the resulting perturbation in the reconstructed donor outcomes is controlled in magnitude, with the constant of proportionality γ coming directly from the operator norm bound in Assumption (A4). □

Lemma 1.3. *Under (A3), for horizon* $k > 0$,

$$|\ell(T_0+k) - \ell^{SHC}(T_0+k)| \ \le \ b_\epsilon(H,k) \ := \ \frac{2\epsilon|k|^{H+1}}{(H+1)!}.$$

Proof of Lemma 1.3. See Proposition 1 of Chen et al. [2024]. □

Remark 1.3. *Lemma 1.3 formalizes the idea that the SHC predictor inherits the smoothness of the latent trend* $\ell(\cdot)$*. Because SHC is built from overlapping donor segments that reflect the same underlying smooth dynamics, the approximation reproduces the first* H *terms of the Taylor expansion around* T_0*. The only discrepancy comes from the* $(H+1)$*-st order remainder, which is small when the latent trend is sufficiently smooth. Thus, the difference between* ℓ *and* ℓ^{SHC} *grows only polynomially in the forecast horizon* k*, and the error is of order* $|k|^{H+1}$ *rather than linear or quadratic in* k*. This ensures that SHC extrapolates the smooth component accurately over short to moderate horizons.*

Lemma 1.4 (Post-treatment smoothness bias and noise)**.** *Under Assumptions (A2) and (A3), the approximation error in the post-treatment period satisfies*

$$\|\mathbf{y}_{post} - \mathbf{Y}_{post}\mathbf{w}^*\| \le b_{\epsilon,post}(H,k) + \delta_{post},$$

where $b_{\epsilon,post}(H,k)$ *is a smoothness bias term similar to the pre-treatment case, and* δ_{post} *bounds the noise in the post-treatment outcomes.*

Proof of Lemma 1.4. Recall from Assumption (A1) that the treated unit's latent potential outcome $\ell(t)$ is $H+1$ times differentiable, and its $(H+1)$-th derivative is bounded by ϵ. By a Taylor expansion argument, the latent outcome over the post-treatment period can be approximated by the linear combination of donor segments with a bias bounded by the smoothness term $b_{\epsilon,\text{post}}(H,k)$.

Additionally, from Assumption (A2), the noise vector $\boldsymbol{\eta}_{\text{post}}$ satisfies $\|\boldsymbol{\eta}_{\text{post}}\| \leq \delta_{\text{post}}$.

Thus,

$$\mathbf{y}_{\text{post}} = \mathbf{Y}_{\text{post}}\mathbf{w}^* + \boldsymbol{\eta}_{\text{post}} + \text{smoothness bias},$$

where the smoothness bias norm is at most $b_{\epsilon,\text{post}}(H,k)$.

Applying the triangle inequality,

$$\|\mathbf{y}_{\text{post}} - \mathbf{Y}_{\text{post}}\mathbf{w}^*\| \leq \|\boldsymbol{\eta}_{\text{post}}\| + b_{\epsilon,\text{post}}(H,k) \leq \delta_{\text{post}} + b_{\epsilon,\text{post}}(H,k).$$

This completes the proof. □

Lemma 1.5 (Weight deviation between oracle and SHC)**.** *Under Assumptions (A1)–(A5), the difference between the oracle weights* $\mathbf{w}^*$ *and the SHC weights* $\mathbf{w}^{SHC}$ *is bounded by*

$$\|\mathbf{w}^* - \mathbf{w}^{SHC}\| \leq \eta,$$

for some $\eta > 0$ *depending on the data and model.*

Proof of Lemma 1.5. This follows from the stability and consistency properties of the SHC estimator under Assumptions (A1)–(A5). Since $\mathbf{w}^{\text{SHC}}$ solves a constrained least squares problem minimizing

$$\left\|\widetilde{\mathbf{y}}_{\text{eval}} - \widetilde{\mathbf{Y}}_{\text{eval}}\mathbf{w}\right\|^2 + \varsigma\|\mathbf{C}_0^\top\mathbf{w}\|^2,$$

and $\mathbf{w}^*$ satisfies the noiseless model $\widetilde{\mathbf{y}}_{\text{eval}} = \widetilde{\mathbf{Y}}_{\text{eval}}\mathbf{w}^* + \boldsymbol{\eta}$ with $\|\boldsymbol{\eta}\| \leq \delta$, standard results on constrained least squares imply $\mathbf{w}^{\text{SHC}}$ is close to $\mathbf{w}^*$ within a ball whose radius depends on δ and model parameters.

More explicitly, since the problem is strictly convex and well-conditioned by assumption (due to the regularization and eigenvalue structure),

$$\|\mathbf{w}^* - \mathbf{w}^{\text{SHC}}\| \leq \eta,$$

where η depends on the noise bound δ, the regularization parameter ς, and the spectral properties of $\widetilde{\mathbf{Y}}_{\text{eval}}$.

A detailed exact expression can be derived by comparing the KKT conditions for $\mathbf{w}^*$ and $\mathbf{w}^{\text{SHC}}$, but for the purposes here, it suffices to assert the existence of such η. □

Lemma 1.6 (Total weight deviation bound)**.** *We have*

$$\|\mathbf{w}^* - \mathbf{w}^{ASHC}\| \leq \|\mathbf{w}^* - \mathbf{w}^{SHC}\| + \|\mathbf{w}^{SHC} - \mathbf{w}^{ASHC}\|.$$

Proof of Lemma 1.6. This is a direct application of the triangle inequality for vector norms. For any vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ in a normed vector space,

$$\|\mathbf{a} - \mathbf{c}\| \leq \|\mathbf{a} - \mathbf{b}\| + \|\mathbf{b} - \mathbf{c}\|.$$

Setting $\mathbf{a} = \mathbf{w}^*$, $\mathbf{b} = \mathbf{w}^{\text{SHC}}$, and $\mathbf{c} = \mathbf{w}^{\text{ASHC}}$ yields the stated result. □

Proof of Theorem 3.1. We want to control the error of the ASHC estimator, given by

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\|.$$

Directly analyzing this is difficult because the weights $\mathbf{w}^{\mathrm{ASHC}}$ are defined through a penalized optimization problem. To make progress, we first recall the triangle inequality, which states that for any vectors $\mathbf{a}$ and $\mathbf{b}$,

$$\|\mathbf{a} + \mathbf{b}\| \leq \|\mathbf{a}\| + \|\mathbf{b}\|.$$

This allows us to introduce the SHC weights into the expression and split the ASHC error into two pieces: the SHC prediction error, which we already know how to bound, and the extra error due to the deviation of ASHC from SHC.

Let $\Delta := \mathbf{w}^{\mathrm{ASHC}} - \mathbf{w}^{\mathrm{SHC}}$ denote the difference between the two weight vectors. Then we can write

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| = \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}} + \widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Applying the triangle inequality gives

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\| + \|\widetilde{\mathbf{Y}}(\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}})\|.$$

Since $\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}} = -\Delta$, the second term becomes $\|\widetilde{\mathbf{Y}}\Delta\|$. Denoting the SHC error by

$$\epsilon_{\mathrm{SHC}} := \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{SHC}}\|,$$

we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\leq\; \epsilon_{\mathrm{SHC}} + \|\widetilde{\mathbf{Y}}\Delta\|.$$

The challenge now reduces to bounding $\|\widetilde{\mathbf{Y}}\Delta\|$. For this, we use the operator norm. Recall that for a matrix A,

$$\|A\| := \sup_{\mathbf{v}\neq 0} \frac{\|A\mathbf{v}\|}{\|\mathbf{v}\|}.$$

This definition guarantees that for any vector $\mathbf{v}$,

$$\|A\mathbf{v}\| \leq \|A\| \cdot \|\mathbf{v}\|.$$

Taking $A = \widetilde{\mathbf{Y}}$ and $\mathbf{v} = \Delta$, we have

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \|\widetilde{\mathbf{Y}}\| \cdot \|\Delta\|.$$

By Assumption (A4), the operator norm of $\widetilde{\mathbf{Y}}$ is bounded by γ, so

$$\|\widetilde{\mathbf{Y}}\Delta\| \leq \gamma\|\Delta\|.$$

To proceed, we need a bound on the size of the deviation Δ. Lemma 3.1 provides exactly this, showing that

$$\|\Delta\| \leq \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Substituting this into the previous inequality, we obtain

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}.$$

Finally, we incorporate the other two sources of error. Lemma 1.3 bounds the bias due to imperfect smoothness of the latent trend by

$$b_\epsilon(H, k) = \frac{2\epsilon|k|^{H+1}}{(H+1)!},$$

and Assumption (A2) ensures that the noise term contributes at most δ. Adding these yields

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} + b_\epsilon(H,k) + \delta.$$

It remains only to factor out ϵ_{SHC} from the first two terms:

$$\epsilon_{\mathrm{SHC}} + \gamma \cdot \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2} = \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right).$$

Thus we conclude that

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}\mathbf{w}^{\mathrm{ASHC}}\| \;\le\; \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right) + b_\epsilon(H,k) + \delta,$$

which completes the proof. □

Proof of Proposition 3.1. We want to bound the out-of-sample prediction error

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\|. \tag{1.60}$$

Step 1: Add and subtract the oracle predictor. Insert $\mathbf{Y}_{\mathrm{post}}\mathbf{w}^*$ inside the norm:

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\| = \left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^* + \mathbf{Y}_{\mathrm{post}}(\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}})\right\|. \tag{1.61}$$

Step 2: Apply the triangle inequality.

$$\le \underbrace{\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^*\right\|}_{\text{smoothness bias + noise}} + \underbrace{\left\|\mathbf{Y}_{\mathrm{post}}(\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}})\right\|}_{\text{weight deviation effect}}. \tag{1.62}$$

Step 3: Bound the smoothness + noise term. By Lemma 1.4,

$$\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^*\| \le b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}}. \tag{1.63}$$

Step 4: Bound the weight deviation effect. By the operator norm bound in Assumption 3.5,

$$\left\|\mathbf{Y}_{\mathrm{post}}(\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}})\right\| \le \gamma_{\mathrm{post}}\left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right\|. \tag{1.64}$$

Using Lemma 1.6,

$$\left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right\| \le \left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{SHC}}\right\| + \left\|\mathbf{w}^{\mathrm{SHC}} - \mathbf{w}^{\mathrm{ASHC}}\right\|. \tag{1.65}$$

By Lemma 1.5 and Lemma 3.1,

$$\left\|\mathbf{w}^* - \mathbf{w}^{\mathrm{ASHC}}\right\| \le \eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}. \tag{1.66}$$

Step 5: Combine bounds.

$$\left\|\mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}}\mathbf{w}^{\mathrm{ASHC}}\right\| \le b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}} + \gamma_{\mathrm{post}}\left(\eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}\right). \tag{1.67}$$

Step 6: Relate prediction error to ATT error. Recall that

$$\widehat{\mathrm{ATT}} - \mathrm{ATT} = \frac{1}{n}\sum_{t\in\mathcal{T}_2}\left(y_t(0) - \hat{y}_t(0)\right). \tag{1.68}$$

(a) Jensen's inequality. For any reals $a_1, \dots, a_n$,

$$\left| \frac{1}{n} \sum_{t=1}^{n} a_t \right| \le \frac{1}{n} \sum_{t=1}^{n} |a_t|.$$

Applying this with $a_t = y_t(0) - \hat{y}_t(0)$ gives

$$\left| \widehat{\mathrm{ATT}} - \mathrm{ATT} \right| \le \frac{1}{n} \sum_{t \in \mathcal{T}_2} \left| y_t(0) - \hat{y}_t(0) \right|. \tag{1.69}$$

(b) Cauchy–Schwarz inequality. Writing the sum as an inner product with $\mathbf{1}$,

$$\frac{1}{n} \sum_{t=1}^{n} |a_t| = \frac{1}{n} \langle |a|, \mathbf{1} \rangle \ \le \ \frac{1}{n} \| \, |a| \, \| \cdot \|\mathbf{1}\|,$$

where $\| \, |a| \, \| = \sqrt{\sum a_t^2}$ and $\|\mathbf{1}\| = \sqrt{n}$. Thus,

$$\frac{1}{n} \sum_{t=1}^{n} |a_t| \le \frac{1}{\sqrt{n}} \sqrt{\sum_{t=1}^{n} a_t^2}. \tag{1.70}$$

Step 7: Express in vector norm form.

$$\left| \widehat{\mathrm{ATT}} - \mathrm{ATT} \right| \le \frac{1}{\sqrt{n}} \left\| \mathbf{y}_{\mathrm{post}} - \mathbf{Y}_{\mathrm{post}} \mathbf{w}^{\mathrm{ASHC}} \right\|. \tag{1.71}$$

Final Step: Substitute the prediction error bound.

$$\left| \widehat{\mathrm{ATT}} - \mathrm{ATT} \right| \le \frac{1}{\sqrt{n}} \left[b_{\epsilon,\mathrm{post}}(H,k) + \delta_{\mathrm{post}} + \gamma_{\mathrm{post}} \left(\eta + \frac{\epsilon_{\mathrm{SHC}}}{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2} \right) \right]. \tag{1.72}$$

□

Proof of Theorem 3.2. Starting from Theorem 3.1, we have

$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}} w^{\mathrm{ASHC}}\| \le \epsilon_{\mathrm{SHC}} \left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}} \right) + b_\epsilon(H,k) + \delta.$$

Step 1. Behavior of ϵ_{SHC}. By (A3), $\ell(t)$ is $(H+1)$-times differentiable. Gaussian kernel smoothing in SHC ensures

$$\epsilon_{\mathrm{SHC}} \to 0 \quad \text{as } m \to \infty, \, H \to \infty.$$

Step 2. Behavior of ASHC penalty term. Since $\epsilon_{\mathrm{SHC}} \to 0$, the inflation factor

$$\frac{\gamma}{\sqrt{2\lambda + 2\varsigma \|\mathbf{C}_0\|^2}} \, \epsilon_{\mathrm{SHC}}$$

vanishes asymptotically.

Step 3. Behavior of smoothness bias $b_\epsilon(H,k)$. From Taylor's theorem,

$$\ell(T_0 + k) = \sum_{h=0}^{H} \frac{\ell^{(h)}(T_0)}{h!} k^h + R_{H+1}(k),$$

with
$$|R_{H+1}(k)| \leq \frac{\epsilon}{(H+1)!}|k|^{H+1}.$$
Thus
$$b_\epsilon(H,k) \leq \frac{2\epsilon}{(H+1)!}|k|^{H+1} \to 0 \quad \text{as } H \to \infty.$$

Step 4. Behavior of noise term. By (A2), $\|\boldsymbol{\eta}\| \leq \delta$. If $\boldsymbol{\eta}$ is i.i.d. sub-Gaussian with variance proxy σ^2, then
$$\frac{\|\boldsymbol{\eta}\|}{m} \xrightarrow{p} 0.$$

Step 5. Combine bounds. Therefore,
$$\limsup_{m\to\infty} \|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\mathrm{ASHC}}\| \leq \delta.$$
With sub-Gaussian noise, the limit is 0 in probability. □

Proof of Corollary 3.1. Part 1: Consistency. From Theorem 3.2, the error converges to δ, or 0 in probability under sub-Gaussian noise.

Part 2: Convergence Rate. From Theorem 3.1, we have
$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\mathrm{ASHC}}\| \leq \epsilon_{\mathrm{SHC}}\left(1 + \frac{\gamma}{\sqrt{2\lambda + 2\varsigma\|\mathbf{C}_0\|^2}}\right) + b_\epsilon(H,k) + \delta.$$

Step 1. Bias term. From Lemma 1.3,
$$b_\epsilon(H,k) = O(h_m^H).$$

Step 2. Stochastic error term. With sub-Gaussian noise, the smoothed variance is
$$O_p\big((mh_m)^{-1/2}\big).$$

Step 3. SHC fit error. Combining,
$$\epsilon_{\mathrm{SHC}} = O_p\big(h_m^H + (mh_m)^{-1/2}\big).$$

Step 4. Optimal bandwidth. Equating orders $h_m^H \asymp (mh_m)^{-1/2}$ yields
$$h_m \asymp m^{-1/(2H+3)}.$$

Step 5. Convergence rate. Substituting,
$$\epsilon_{\mathrm{SHC}} = O_p\big(m^{-H/(2H+3)}\big).$$
Thus the total error is
$$\|\widetilde{\mathbf{y}} - \widetilde{\mathbf{Y}}w^{\mathrm{ASHC}}\| = O_p\big(m^{-H/(2H+3)}\big).$$

Step 6. Near-parametric rate. As $H \to \infty$,
$$\frac{H}{2H+3} \to \tfrac{1}{2}.$$
So the rate approaches $O_p(m^{-1/2})$. □

Remark 1.4. *The deterministic noise bound δ in (A2) corresponds to $\|\boldsymbol{\eta}\| \leq \delta$. Under sub-Gaussian noise, $\delta = O(\sigma\sqrt{m})$, while kernel smoothing reduces effective noise to $O_p((mh_m)^{-1/2})$. This explains the appearance of the nonparametric rate.*

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